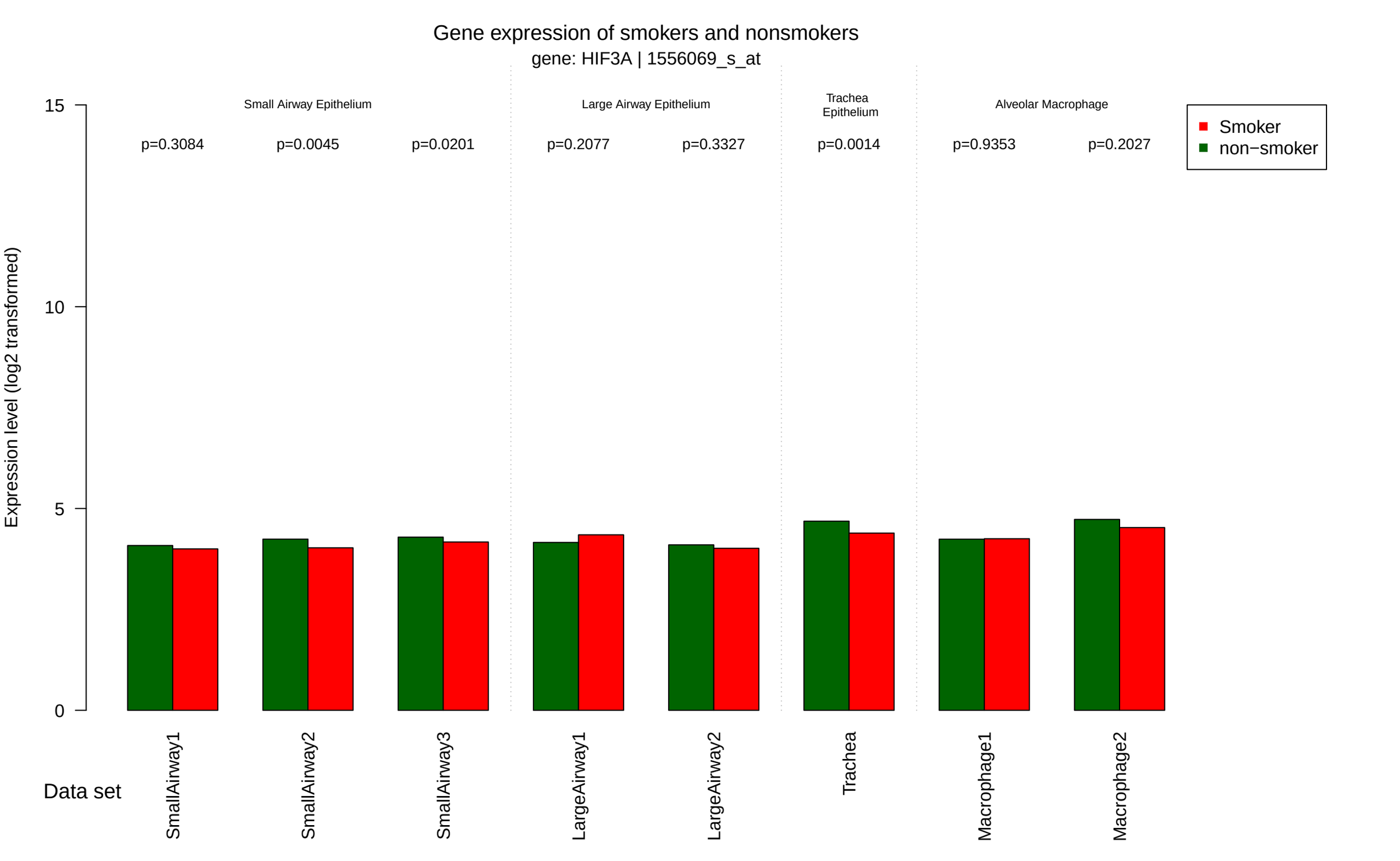
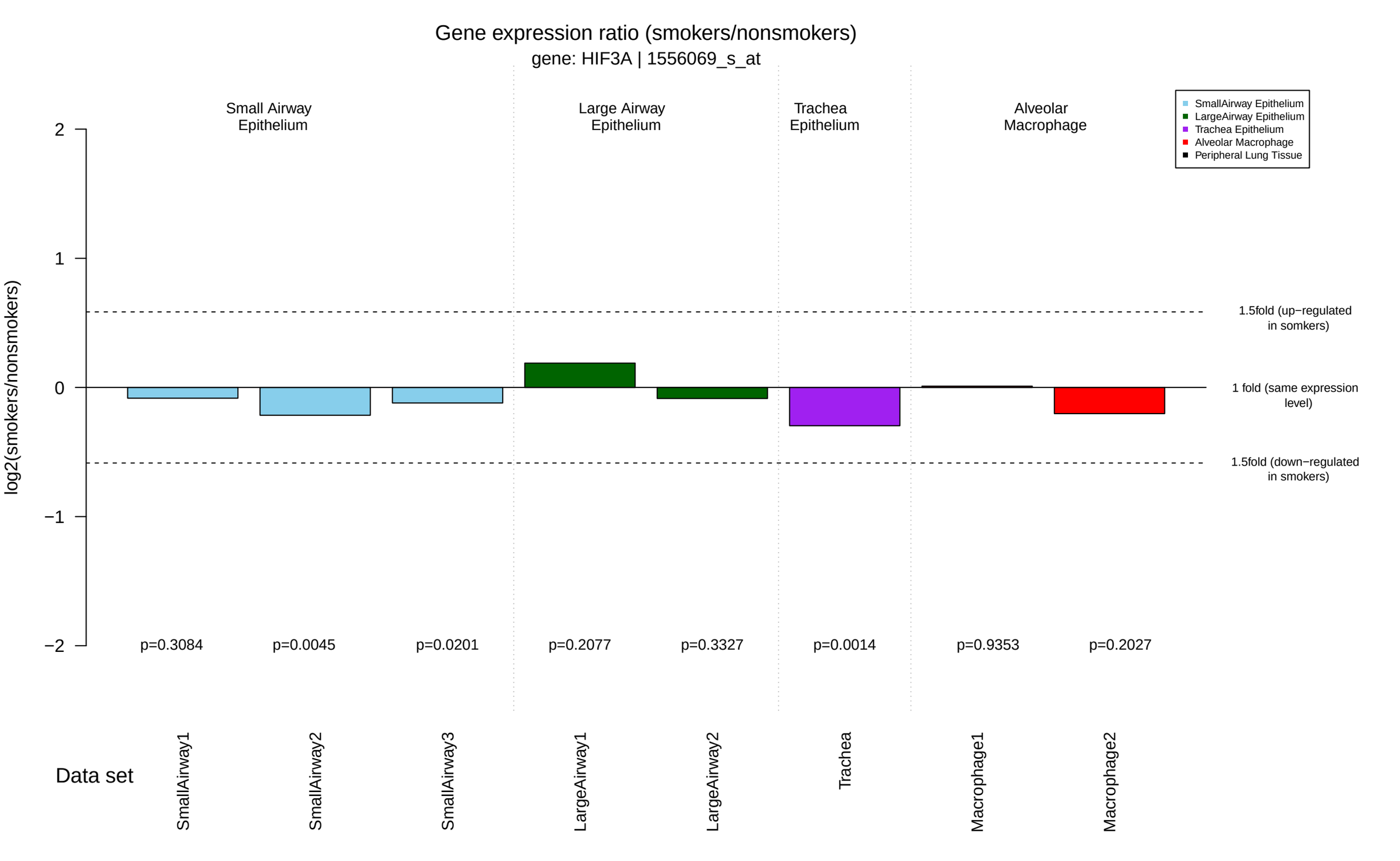
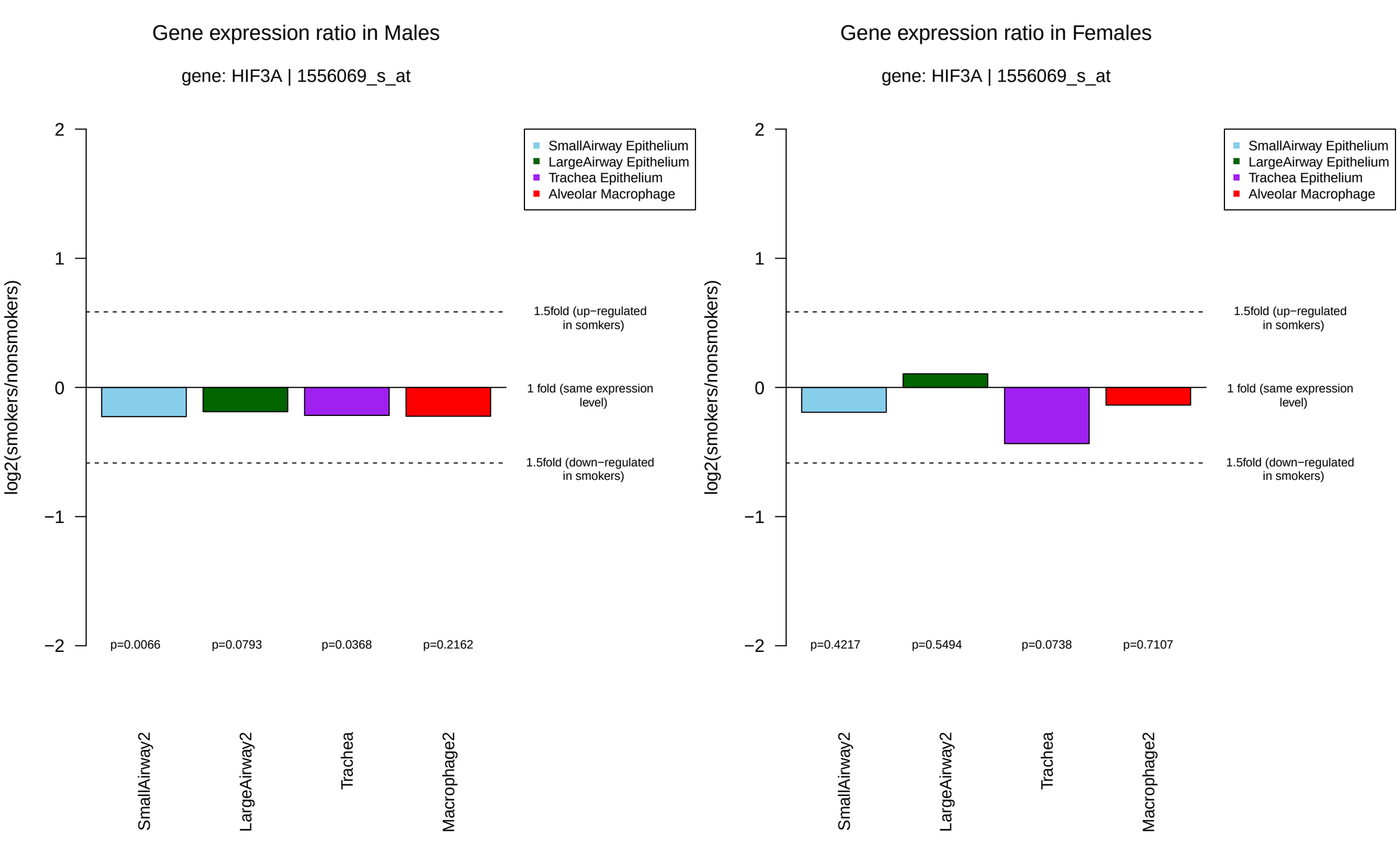
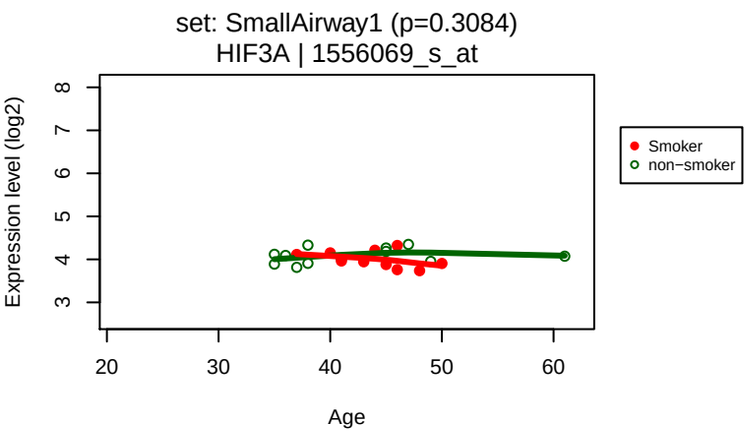
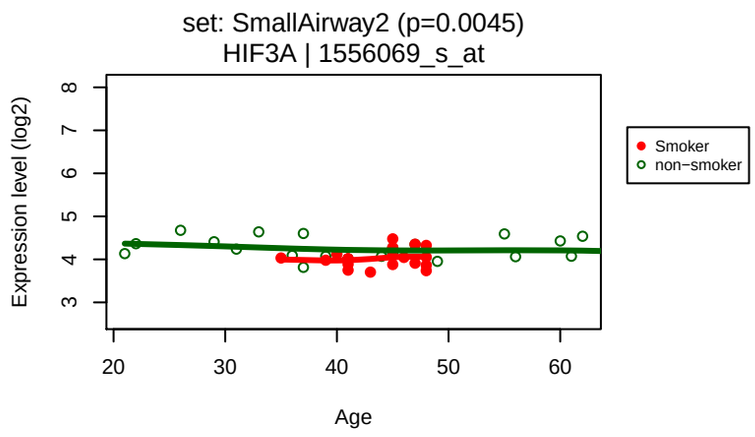
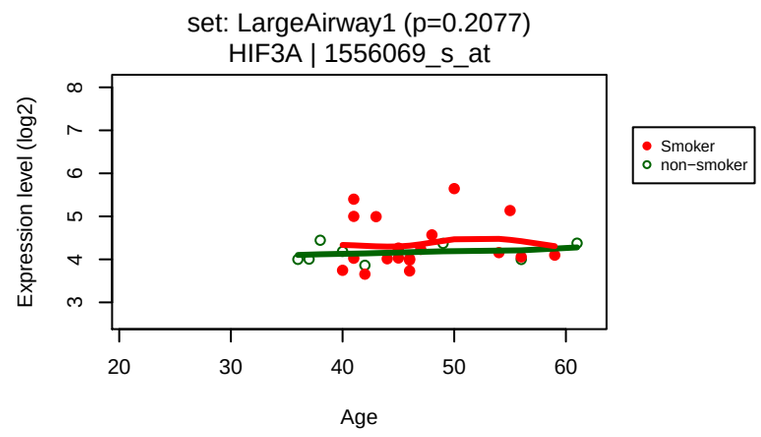

Gene expression of smokers and nonsmokers
gene: HIF3A | 1556069_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3084
p=0.0045
p=0.0201
p=0.2077
p=0.3327
p=0.0014
p=0.9353
p=0.2027
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

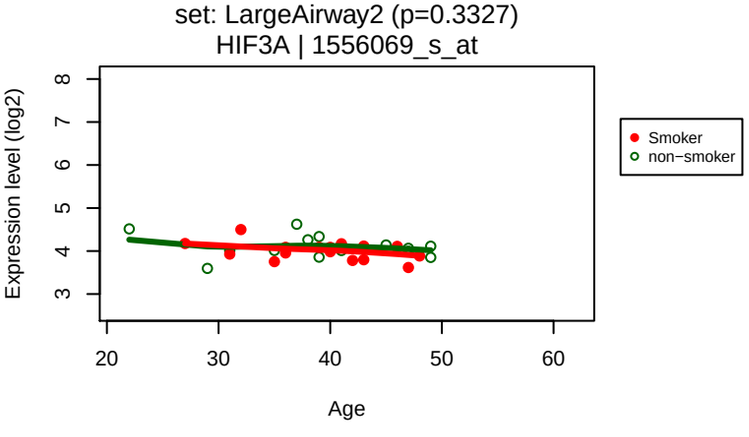

Gene expression ratio (smokers/nonsmokers)
gene: HIF3A | 1556069_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3084
p=0.0045
p=0.0201
p=0.2077
p=0.3327
p=0.0014
p=0.9353
p=0.2027
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

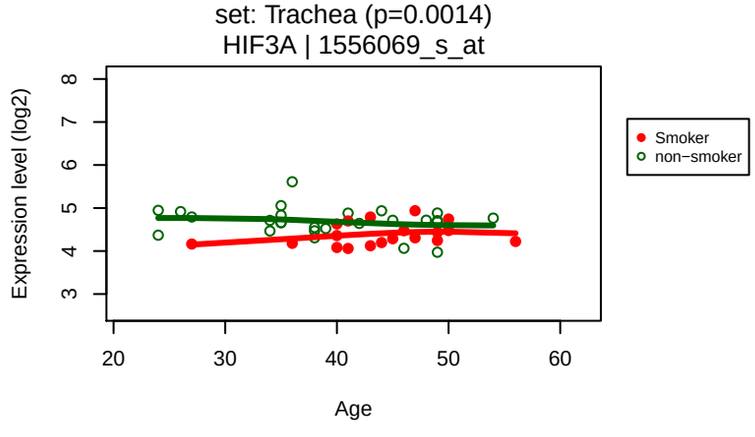

Gene expression ratio in Males
gene: HIF3A | 1556069_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0066
p=0.0793
p=0.0368
p=0.2162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIF3A | 1556069_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4217
p=0.5494
p=0.0738
p=0.7107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

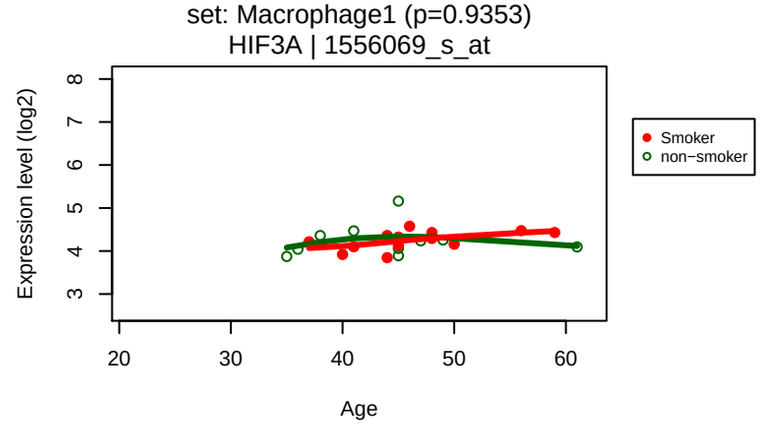

set: SmallAirway1 (p=0.3084)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker

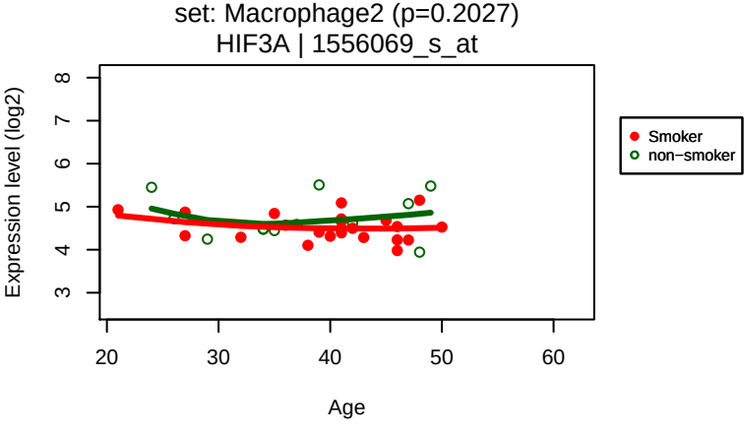

set: SmallAirway2 (p=0.0045)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker

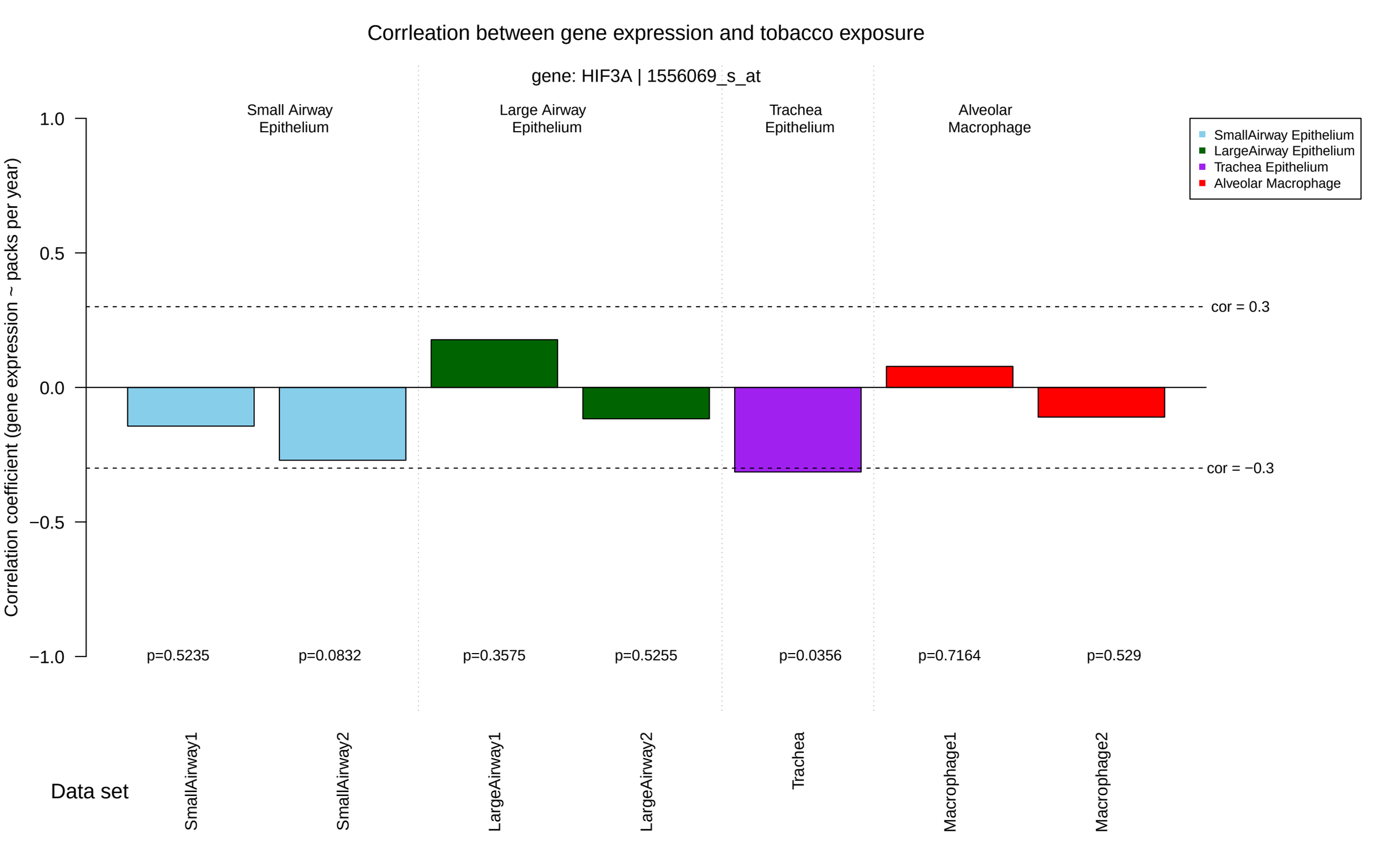

set: LargeAirway1 (p=0.2077)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3327)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9353)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2027)
HIF3A | 1556069_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIF3A | 1556069_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5235
p=0.0832
p=0.3575
p=0.5255
p=0.0356
p=0.7164
p=0.529
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2